
Software Requirements Specification For

FRA Atlas: A WebGIS-Based Intelligent Frontier for Forest Rights Decision Support

Version 1.0 approved

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the requirements for the system titled “**Development of FRA Atlas and WebGIS-Based Decision Support System (DSS) for Integrated Monitoring of Forest Rights Act (FRA) Implementation**”, version 1.0.

The purpose of this system is to digitize and manage FRA records, integrate them with spatial data, and provide an intelligent platform for monitoring and decision-making. The system uses AI-based technologies such as OCR and Named Entity Recognition (NER) to convert scanned FRA documents into structured digital data, along with a WebGIS interface to visualize land records and assets. It also includes a Decision Support System (DSS) for recommending suitable government schemes and a blockchain-based layer to ensure data integrity and prevent tampering of approved records.

This SRS covers the complete system including document digitization, data extraction, spatial visualization, asset classification, decision support, and secure data management.

1.2 Document Conventions

This document follows standard IEEE SRS conventions. The following notations and conventions are used:

- “**Shall**” indicates a mandatory requirement that must be implemented.
- “**Should**” indicates a recommended requirement.
- “**May**” indicates an optional feature.
- Requirements are uniquely identified using the format **REQ-X** (e.g., REQ-1, REQ-2).
- Headings and subheadings are numbered hierarchically for clarity.
- Technical terms such as OCR, NER, CNN, DSS, and WebGIS are used consistently throughout the document.
- Priorities for requirements are categorized as **High, Medium, or Low**, and each feature explicitly specifies its priority.
- All data flows and processes described are aligned with the system architecture and workflow of the project.

1.3 Intended Audience and Reading Suggestions

This SRS document is intended for multiple stakeholders involved in the development and use of the system:

- **Developers** – To understand system architecture, modules, and implementation requirements.
- **Testers** – To design and execute test cases based on functional and non-functional requirements.
- **Project Evaluators / Faculty** – To evaluate system design, feasibility, and completeness.
- **Forest Officers (Admin Users)** – To understand document upload, verification, and approval workflows.
- **Public Users** – To understand how to view FRA data and track claim status.
- **Project Managers** – To monitor project scope, progress, and deliverables.

1.3.1 Reading Suggestions:

- Start with the **Introduction** to understand the system purpose and scope.
- Read the **Overall Description** for a high-level overview of system components and user roles.
- Refer to the **System Features** section for detailed functionalities such as OCR, NER, WebGIS, DSS, and Blockchain.
- Review **External Interface Requirements** for understanding system interactions.
- Finally, go through **Non-functional Requirements** for performance, security, and quality aspects.

1.4 Product Scope

The **FRA Atlas: A WebGIS-Based Intelligent Frontier for Forest Rights Decision Support** is designed to provide a comprehensive digital platform for managing and monitoring the implementation of the Forest Rights Act (FRA), 2006.

The system aims to address key challenges such as non-digitized records, lack of spatial integration, and absence of decision support tools. It achieves this by:

- Digitizing FRA documents using AI-based OCR and extracting structured data using NER
- Visualizing FRA land records on an interactive WebGIS platform
- Identifying land-use and assets using AI-based CNN models
- Providing a Decision Support System (DSS) to recommend government schemes

- Ensuring data integrity and transparency using a blockchain-based mechanism

The system supports both **administrative users (forest officers)** and **public users**, improving transparency, efficiency, and accessibility of FRA data.

This project aligns with government objectives of digital governance, data-driven decision-making, and effective implementation of welfare schemes for forest-dependent communities.

1.5 References

The following documents and resources are referred to in this SRS:

1. Forest Rights Act (FRA), 2006 Government of India – Ministry of Tribal Affairs Available at: <https://tribal.nic.in/FRA.aspx>
2. Problem Statement ID 25108 – Ministry of Tribal Affairs (MoTA) “Development of FRA Atlas and WebGIS-Based Decision Support System”
3. PM-JGUA (DHARTI AABA JATIYA GRAM UTKARSH ABHIYAN) Guidelines Ministry of Tribal Affairs
4. Monthly Progress Reports (MPR) on FRA Implementation Ministry of Tribal Affairs
5. ISRO Bhuvan Geoportal Documentation <https://bhuvan.nrsc.gov.in>
6. Tesseract OCR Documentation <https://github.com/tesseract-ocr/tesseract>
7. TensorFlow/Keras Documentation (CNN Models) <https://www.tensorflow.org>
8. LangChain and Google Gemini API Documentation <https://python.langchain.com>

2. Overall Description

2.1 Product Perspective

The FRA Atlas: A WebGIS-Based Intelligent Frontier for Forest Rights Decision Support is a new, self-contained web-based system developed to address the challenges in the implementation of the Forest Rights Act (FRA), 2006.

Currently, FRA data is maintained in fragmented, non-digitized formats and lacks integration with spatial and decision-support systems. This product serves as an integrated platform that combines:

- AI-based document digitization (OCR + NER)

- Spatial data visualization (WebGIS)
- Asset classification (CNN)
- Decision support (DSS using LLM)
- Data integrity (Blockchain)

The system interacts with external components such as:

- ISRO Bhuvan API (for satellite maps)
- Gemini API (for DSS recommendations)
- OCR engines (Tesseract)

Thus, it acts as a centralized intelligent system bridging data, GIS, and decision-making.

2.2 Product Functions

- The system provides the following major functionalities:
- Upload and digitize FRA documents
- Extract structured data using OCR and NER
- Store FRA data in a spatial database
- Allow officers to verify and manage records
- Enable manual drawing or integration of land boundaries
- Classify land-use assets using CNN
- Display FRA data on an interactive WebGIS map
- Provide filtering by state, district, and village
- Generate scheme recommendations using DSS (LangChain + Gemini)
- Ensure data integrity using blockchain-based hashing
- Provide role-based access (Officer and Public user)

2.3 User Classes and Characteristics

The various users are

2.3.1 Forest Officer (Primary User)

- Role: Admin / Authorized user
- Responsibilities: Upload, verify, approve FRA records
- Technical Level: Moderate
- Access Level: Full access (read/write)

2.3.2 Public User (Secondary User)

- Role: Viewer
- Responsibilities: View FRA data and track claims
- Technical Level: Basic
- Access Level: Read-only

2.3.3 System Administrator (Optional)

- Role: Maintain system
- Responsibilities: Manage database, monitor system
- Technical Level: High
- Access Level: Full system control

2.4 Operating Environment

The system operates in a web-based environment:

- Frontend: React (runs on web browsers like Chrome, Edge)
- Backend: FastAPI (Python-based server)
- Database: PostgreSQL with PostGIS
- AI/ML:
 - OCR: Tesseract
 - CNN: TensorFlow/Keras
 - DSS: LangChain + Gemini

- GIS: ISRO Bhuvan API, Mapbox/Leaflet
- Operating Systems: Windows / Linux
- Deployment: Docker-based environment

2.5 Design and Implementation Constraints

- Dependence on OCR accuracy for correct data extraction
- Availability of satellite data via Bhuvan API
- Limited labeled datasets for CNN training
- Internet connectivity required for API communication
- Use of specific technologies: FastAPI, React, PostgreSQL, PostGIS
- Security requirements including JWT authentication and blockchain integration
- Compliance with government data handling standards

2.6 User Documentation

The system will provide:

- User Manual (for Forest Officers and Public Users)
- Installation Guide (for developers/admins)
- Online Help / Tooltips within the application
- Technical Documentation for system modules
- Documentation will be provided in:
- PDF format
- Web-based help interface

2.7 Assumptions and Dependencies

Assumptions:

- FRA documents are available in scanned format
- Users have access to internet-enabled devices
- Officers are trained to verify and input data
- OCR can extract sufficient readable text

Dependencies:

- ISRO Bhuvan API for satellite imagery
- Gemini API for DSS functionality
- Tesseract OCR for text extraction
- PostgreSQL/PostGIS for spatial data storage
- External datasets (FRA records, policy data)

The system's performance and accuracy depend on the reliability of these external components.

3. External Interface Requirements

3.1 User Interfaces

The system provides a web-based graphical user interface (GUI) accessible through modern browsers.

Main Interfaces:

1. Login Interface
 - Username and password input
 - Role-based login (Officer / Public)
 - Error messages for invalid credentials
2. Forest Officer Dashboard
 - Upload FRA documents (PDF/Image)
 - View extracted OCR + NER data
 - Verify and approve records

- Draw land boundaries on map
- 3. Public User Interface
 - View FRA records
 - Search/filter by State, District, Village
 - Track claim status
- 4. WebGIS Interface
 - Interactive map display
 - Satellite layer (Bhuvan)
 - FRA land boundaries overlay
 - Clickable polygons showing:
 - OCR data
 - CNN asset type
 - DSS recommendations

UI Characteristics:

- Responsive design (desktop/laptop)
- Consistent layout using React + TailwindCSS
- Standard buttons: Submit, Upload, Filter, View
- Error handling:
 - Invalid input → error message displayed
 - Missing fields → validation prompts

3.2 Hardware Interfaces

The system does not require specialized hardware.

Supported devices:

- Desktop / Laptop
- Internet-enabled devices

Hardware interaction:

- Input via keyboard and mouse
- Display through standard monitor
- Optional: GPS or GIS data sources (external systems)

3.3 Software Interfaces

The system interacts with the following software components:

Backend Framework FastAPI (REST APIs)

Database: PostgreSQL + PostGIS (spatial data storage)

AI Components:

OCR: Tesseract

CNN: TensorFlow/Keras

DSS: LangChain + Gemini API

GIS Integration: ISRO Bhuvan API (WMS services)

3.4 Communications Interfaces

The system uses standard web communication protocols:

Protocol: HTTP / HTTPS

API Type: REST APIs

Data Format: JSON

External Communications:

Bhuvan API → map tiles (WMS requests)

Gemini API → DSS responses

Security:

JWT-based authentication

Encrypted communication (HTTPS)

4. System Features

4.1 FRA Document Digitization (OCR + NER)

4.1.1 Description and Priority

This feature enables digitization of scanned FRA documents using OCR and extraction of structured data using NER.

Priority: High

4.1.2 Stimulus/Response Sequences

User uploads document → System processes using OCR → Extracted text displayed → NER extracts structured data → Data stored in database

4.1.3 Functional Requirements

REQ-1: System shall allow uploading of FRA documents (PDF/Image)

REQ-2: System shall extract text using OCR

REQ-3: System shall extract entities (name, village, claim type, status)

REQ-4: System shall store extracted data in structured format

REQ-5: System shall handle OCR errors and notify user

4.2 Asset Classification (CNN)

4.2.1 Description and Priority

This feature classifies land-use types using a trained CNN model.

Priority: High

4.2.2 Stimulus/Response Sequences

Land image is provided → Preprocessing applied → CNN model predicts class → Output stored and displayed

4.2.3 Functional Requirements

REQ-6: System shall accept satellite/land images

REQ-7: System shall preprocess images

REQ-8: System shall classify land into categories

REQ-9: System shall store classification result

4.3 WebGIS Visualization

4.3.1 Description and Priority

Displays FRA records and assets on an interactive map.

Priority: High

4.3.2 Stimulus/Response Sequences

User opens map → Bhuvan layer loads → FRA data overlays → User clicks polygon → Popup displays details

4.3.3 Functional Requirements

REQ-10: System shall load satellite maps using Bhuvan API

REQ-11: System shall display FRA spatial data

REQ-12: System shall allow user interaction

REQ-13: System shall show popup with details

4.4 Decision Support System (DSS)

4.4.1 Description and Priority

Provides scheme recommendations using LangChain and Gemini.

Priority: High

4.4.2 Stimulus/Response Sequences

System collects FRA + asset data → LangChain builds prompt → Gemini generates response → Output displayed

4.4.3 Functional Requirements

REQ-14: System shall generate prompts using structured data

REQ-15: System shall send data to Gemini API

REQ-16: System shall generate recommendations

REQ-17: System shall display reasoning

4.5 Blockchain Integrity

4.5.1 Description and Priority

Ensures data integrity using hash-based blockchain mechanism.

Priority: Medium

4.5.2 Stimulus/Response Sequences

Record approved → Hash generated → Stored securely → On access → verified

4.5.3 Functional Requirements

REQ-18: System shall generate hash for records

REQ-19: System shall store hash

REQ-20: System shall verify data integrity

4.6 Authentication and Authorization

4.6.1 Description and Priority

Provides secure access using JWT and role-based access.

Priority: High

4.6.2 Stimulus/Response Sequences

User logs in → JWT generated → Request validated → Access granted/restricted

4.6.3 Functional Requirements

REQ-21: System shall authenticate users

REQ-22: System shall implement role-based access

REQ-23: System shall restrict unauthorized access

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- The system shall respond to user requests within **2–5 seconds** under normal load conditions.
- OCR processing of uploaded FRA documents shall complete within an acceptable time (typically **5–10 seconds per document**, depending on size).
- WebGIS map loading and layer rendering shall occur within **3–5 seconds** with stable internet connectivity.
- The system shall support **multiple concurrent users** (at least 50 users) without significant performance degradation.

- Database queries for FRA records and spatial data shall be optimized to ensure fast retrieval.

Rationale:

Efficient performance ensures smooth user experience for officers and public users, especially during data access and decision-making.

5.2 Safety Requirements

- The system shall prevent accidental deletion or modification of FRA records by implementing validation and confirmation steps.
- Approved FRA records shall not be modified without proper authorization.
- Input validation shall be applied to prevent incorrect or incomplete data entry.
- The system shall maintain backup mechanisms to prevent data loss.

Rationale:

Ensures protection of critical land ownership data and prevents errors that may affect decision-making.

5.3 Security Requirements

- The system shall implement **JWT-based authentication** for all users.
- Role-Based Access Control (RBAC) shall be enforced:
 - Officers → Upload, verify, approve
 - Public users → View only
- All communication shall occur over **secure HTTPS protocol**.
- Sensitive data shall be protected from unauthorized access.
- Blockchain-based hashing shall be used to ensure **data integrity and tamper detection**.
- The system shall log user activities for auditing purposes.

Rationale:

Ensures secure handling of sensitive FRA records and prevents unauthorized modifications.

5.4 Software Quality Attributes

- **Usability:**
The system shall provide a simple and user-friendly interface for both technical and non-technical users.
- **Reliability:**
The system shall ensure consistent and correct operation with minimal failures.
- **Maintainability:**
The system shall follow modular architecture for easy updates and maintenance.
- **Scalability:**
The system shall support expansion to additional regions, users, and datasets.
- **Performance Efficiency:**
The system shall handle large datasets and spatial queries efficiently.
- **Interoperability:**
The system shall integrate with external APIs such as Bhuvan and Gemini.
- **Security:**
The system shall ensure confidentiality, integrity, and availability of data.

5.5 Business Rules

- Only **Forest Officers** are authorized to upload, verify, and approve FRA records.
- **Public users** are restricted to viewing data and cannot modify records.
- Once a record is **approved**, it becomes immutable and cannot be altered (enforced using blockchain hashing).
- FRA records must follow standardized formats for consistency across districts.
- DSS recommendations shall be based on available FRA data and predefined policy rules.

6. Other Requirements

The following additional requirements are applicable to the system:

6.1.1 Database Requirements

- The system shall use **PostgreSQL with PostGIS** for storing structured and spatial data.

- The database shall support storage of:
 - FRA records
 - Spatial geometries (polygons)
 - Blockchain hashes
- Data consistency and integrity shall be maintained using constraints and validations.

6.1.2 Legal and Regulatory Requirements

- The system shall comply with guidelines of the **Forest Rights Act (FRA), 2006**.
- User data and sensitive information shall be handled according to applicable data protection policies.

6.1.3 Scalability Requirements

- The system shall support expansion to multiple states, districts, and villages.
- The architecture shall allow integration with additional datasets and government schemes in the future.

6.1.4 Deployment Requirements

- The system shall be deployable using **Docker containers**.
- CI/CD pipelines (e.g., GitHub Actions) shall be used for automated deployment.

6.1.5 Reuse Objectives

- Modular components such as OCR, NER, WebGIS, and DSS shall be reusable in similar government or GIS-based applications.

Appendix A: Glossary

Term	Definition
FRA	Forest Rights Act, 2006
OCR	Optical Character Recognition
NER	Named Entity Recognition
CNN	Convolutional Neural Network
DSS	Decision Support System
WebGIS	Web-based Geographic System

Term	Definition
WMS	Web Map Service
JWT	JSON Web Token
RBAC	Role-Based Access Control
API	Application Programming Interface
PostGIS	Spatial extension of PostgreSQL
LLM	Large Language Model
LangChain	Framework for managing LLM interactions
Gemini	Google's LLM used for DSS

Appendix B: Analysis Models

The following models and diagrams are used in the system:

- System Architecture Diagram
- User Flow Diagram (Forest Officer and Public User)
- Data Flow Diagram (OCR → NER → Database → WebGIS → DSS)
- WebGIS Layer Architecture
- Blockchain Flow (Hash generation and verification)

Appendix C: To Be Determined List

The following items are marked for future clarification:

1. Final dataset size for CNN training
2. Accuracy metrics for OCR and NER performance
3. Exact blockchain implementation (private/public configuration)
4. Integration details for additional government schemes
5. Performance optimization for large-scale deployment